Vendor Performance Analysis

Project Overview

This project focuses on effective inventory and sales management which are critical for optimizing profitability in the retail and wholesale industry. This analysis helps companies ensure that they are not incurring losses due to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to:

- Identify underperforming brands that require promotional or pricing adjustments.
- Determine top vendors contributing to sales and gross profit.
- Analyze the impact of bulk purchasing on unit costs.
- Assess inventory turnover to reduce holding costs and improve efficiency.
- Investigate the profitability variance between high-performing and low-performing vendors.

Dataset Summary

This analysis is done on 6 datasets comprising of interlinked data.

1. **`begin_inventory.csv`**
   a. Rows: 206529
   b. Columns: 9
   c. Table description:

```
 Table Name : begin_inventory
   column_name column_type null   key default extra
0  inventoryid     VARCHAR  YES  None    None  None
1        store      BIGINT  YES  None    None  None
2         city     VARCHAR  YES  None    None  None
3        brand      BIGINT  YES  None    None  None
4  description     VARCHAR  YES  None    None  None
5         size     VARCHAR  YES  None    None  None
6       onhand      BIGINT  YES  None    None  None
7        price      DOUBLE  YES  None    None  None
8    startdate        DATE  YES  None    None  None
```

2. **`end_inventory.csv`**
   a. Rows: 224489
   b. Columns: 9
   c. Table description:

Table Name : end_inventory

	column_name	column_type	null	key	default	extra
0	inventoryid	VARCHAR	YES	None	None	None
1	store	BIGINT	YES	None	None	None
2	city	VARCHAR	YES	None	None	None
3	brand	BIGINT	YES	None	None	None
4	description	VARCHAR	YES	None	None	None
5	size	VARCHAR	YES	None	None	None
6	onhand	BIGINT	YES	None	None	None
7	price	DOUBLE	YES	None	None	None
8	enddate	DATE	YES	None	None	None

3. **`purchase_prices.csv`**
    a. Rows: 12261
    b. Columns: 9
    c. Table description:

Table Name : purchase_prices

	column_name	column_type	null	key	default	extra
0	brand	BIGINT	YES	None	None	None
1	description	VARCHAR	YES	None	None	None
2	price	DOUBLE	YES	None	None	None
3	size	VARCHAR	YES	None	None	None
4	volume	VARCHAR	YES	None	None	None
5	classification	BIGINT	YES	None	None	None
6	purchaseprice	DOUBLE	YES	None	None	None
7	vendornumber	BIGINT	YES	None	None	None
8	vendorname	VARCHAR	YES	None	None	None

4. **`purchases.csv`**
    a. Rows: 2372474
    b. Columns: 16
    c. Tables description:

Table Name : purchases

	column_name	column_type	null	key	default	extra
0	inventoryid	VARCHAR	YES	None	None	None
1	store	BIGINT	YES	None	None	None
2	brand	BIGINT	YES	None	None	None
3	description	VARCHAR	YES	None	None	None
4	size	VARCHAR	YES	None	None	None
5	vendornumber	BIGINT	YES	None	None	None
6	vendorname	VARCHAR	YES	None	None	None
7	ponumber	BIGINT	YES	None	None	None
8	podate	DATE	YES	None	None	None
9	receivingdate	DATE	YES	None	None	None
10	invoicedate	DATE	YES	None	None	None
11	paydate	DATE	YES	None	None	None
12	purchaseprice	DOUBLE	YES	None	None	None
13	quantity	BIGINT	YES	None	None	None
14	dollars	DOUBLE	YES	None	None	None
15	classification	BIGINT	YES	None	None	None

5. **`sales.csv`**

a. Rows: 12825363

b. Columns: 14

c. Table description:

Table Name : sales

	column_name	column_type	null	key	default	extra
0	inventoryid	VARCHAR	YES	None	None	None
1	store	BIGINT	YES	None	None	None
2	brand	BIGINT	YES	None	None	None
3	description	VARCHAR	YES	None	None	None
4	size	VARCHAR	YES	None	None	None
5	salesquantity	BIGINT	YES	None	None	None
6	salesdollars	DOUBLE	YES	None	None	None
7	salesprice	DOUBLE	YES	None	None	None
8	salesdate	DATE	YES	None	None	None
9	volume	DOUBLE	YES	None	None	None
10	classification	BIGINT	YES	None	None	None
11	excisetax	DOUBLE	YES	None	None	None
12	vendorno	BIGINT	YES	None	None	None
13	vendorname	VARCHAR	YES	None	None	None

6. **`vendor_invoice.csv`**

a. Rows: 5543

b. Columns: 10

c. Table description:

Table Name : vendor_invoice

	column_name	column_type	null	key	default	extra
0	vendornumber	BIGINT	YES	None	None	None
1	vendorname	VARCHAR	YES	None	None	None
2	invoicedate	DATE	YES	None	None	None
3	ponumber	BIGINT	YES	None	None	None
4	podate	DATE	YES	None	None	None
5	paydate	DATE	YES	None	None	None
6	quantity	BIGINT	YES	None	None	None
7	dollars	DOUBLE	YES	None	None	None
8	freight	DOUBLE	YES	None	None	None
9	approval	VARCHAR	YES	None	None	None

Data Ingestion and Table creation

Leveraged DUCKDB connection for reading the csv files and then ingested them into database `'my_db.duckdb'` into tables.
DUCKDB ensures batch processing by reading multiple file chunks in parallel and using vectorized execution.

```
...
    ---------------------------------------------------

    Starting: begin_inventory.csv
    Data Ingested to Table: begin_inventory
    Total Ingestion time for begin_inventory : 0.01808757781982422 minutes

    ---------------------------------------------------

    Starting: end_inventory.csv
    Data Ingested to Table: end_inventory
    Total Ingestion time for end_inventory : 0.014534461498260497 minutes

    ---------------------------------------------------

    Starting: purchases.csv

... FloatProgress(value=0.0, layout=Layout(width='auto'), style=ProgressStyle(bar_color='black'))

... Data Ingested to Table: purchases
    Total Ingestion time for purchases : 0.26087273359298707 minutes

    ---------------------------------------------------

    Starting: purchase_prices.csv
    Data Ingested to Table: purchase_prices
    Total Ingestion time for purchase_prices : 0.016794339815775553 minutes

    ---------------------------------------------------

    Starting: sales.csv

... FloatProgress(value=0.0, layout=Layout(width='auto'), style=ProgressStyle(bar_color='black'))

... Data Ingested to Table: sales
    Total Ingestion time for sales : 0.645958399772644 minutes

    ---------------------------------------------------

    Starting: vendor_invoice.csv
    Data Ingested to Table: vendor_invoice
    Total Ingestion time for vendor_invoice : 0.014971478780110677 minutes
```

Exploratory Data Analysis using Python and SQL – ETL Pipeline

Key steps performed:

1. Data cleaning and validation
2. Handling missing values
3. Understanding distributions of key variables
4. Identifying anomalies and outliers
5. Feature creation (e.g., `avg_sales_price`, `gross_profit`, `profit_margin_percent`, `stock_turnover` etc.)

Key Insights:

1. Data was consistent across most fields with minor anomalies
    - Identified same vendor number having different vendor names and fixed the same.

	vendornumber	vendorname	cnt
0	1587	VINEYARD BRANDS INC	55
1	1587	VINEYARD BRANDS LLC	3
2	2000	SOUTHERN WINE & SPIRITS NE	55
3	2000	SOUTHERN GLAZERS W&S OF NE	12
4	4425	MARTIGNETTI COMPANIES	55
5	4425	MARTIGNETTI COMPANIES	27

2. Pricing and quantity distributions were skewed, requiring aggregation
3. Derived metrics enabled deeper vendor-level insights

Data Analysis using SQL (Vendor Performance Analysis)

Approach:

1. Aggregated vendor-level metrics using SQL
   a. `vendor_sale_purchase_summary`
   b. `vendor_freight_summary`
2. Analyzed risk bucket transitions
3. Assessed loan-to-value stability

Key Analysis and Findings:

1. **Brands that need promotional and pricing adjustment**
   Top 10 Brands that need promotional or pricing adjustment which exhibit lower sales performance but higher profit margins Brands lying in < 25%ile in terms of sales and > 75%ile in terms of profit margins are the brands to be targeted.

	brandname	total_sales_dollars	avg_profit_margin_percent
0	M Chiarlo Gavi Wh	1208.90	99.39
1	The Club Strawbry Margarita	143.28	98.97
2	Dom Janasse Terre d'Argile	1019.49	98.68
3	Cecchi Natio Organic Chianti	735.08	98.58
4	Skinnygirl Cucumber Vodka	779.48	98.56
5	Barbarossa Spiced Rum	945.14	98.22
6	Bacardi Oakheart Spiced Trav	399.60	98.13
7	Chi Chi's Chocolate Malt RTD	461.58	98.11
8	Kesselstatt Piesporter Gold	929.69	97.81
9	Penfolds B28 Shiraz/Kalimna	890.73	97.78

2. **Highest Sales Performance**
   a. Vendors that demonstrate high sales and lie in top 5%ile.

	vendorname	total_sales_dollar
0	DIAGEO NORTH AMERICA INC	67990099.42
1	MARTIGNETTI COMPANIES	39276398.80
2	PERNOD RICARD USA	32063196.19
3	JIM BEAM BRANDS COMPANY	31423020.46
4	BACARDI USA INC	24854817.14
5	CONSTELLATION BRANDS INC	24218745.65

b. Brands that demonstrate high sales and lie in top 1%ile.

	brandname	total_sales_dollar
0	Jack Daniels No 7 Black	7964746.76
1	Tito's Handmade Vodka	7399657.58
2	Grey Goose Vodka	7209608.06
3	Capt Morgan Spiced Rum	6356320.62
4	Absolut 80 Proof	6244752.03
...	...	...
73	Zhenka 80 Proof	816412.77
74	Remy Martin VSOP Cognac	815926.80
75	Apothic Winemakers Red Blend	812899.38
76	Seagrams 7 Crown	801560.19
77	Sambuca Romana	763390.62

78 rows × 2 columns

3. **Purchase Contribution**
   Vendors which contribute the most to total purchase contribution

	vendorname	total_purchase_dollars	contribution_perc
0	DIAGEO NORTH AMERICA INC	50097226.16	16.30
1	MARTIGNETTI COMPANIES	25464774.04	8.29
2	PERNOD RICARD USA	23851164.17	7.76
3	JIM BEAM BRANDS COMPANY	23494304.32	7.64
4	BACARDI USA INC	17432020.26	5.67
5	CONSTELLATION BRANDS INC	15273708.08	4.97
6	BROWN-FORMAN CORP	13238939.18	4.31
7	E & J GALLO WINERY	12068539.22	3.93
8	ULTRA BEVERAGE COMPANY LLP	11167081.61	3.63
9	M S WALKER INC	9764312.60	3.18

4. **Order Size Impact**
   Avg. unit price by order bucket to evaluate optimal purchase volume for cost savings.

	ordersize	avg_unit_price
0	Medium	15.48
1	Large	10.78
2	Small	39.08

5. **Inventory Turnover**
   Vendors having low inventory turnover, indicating excess stock and slow moving products

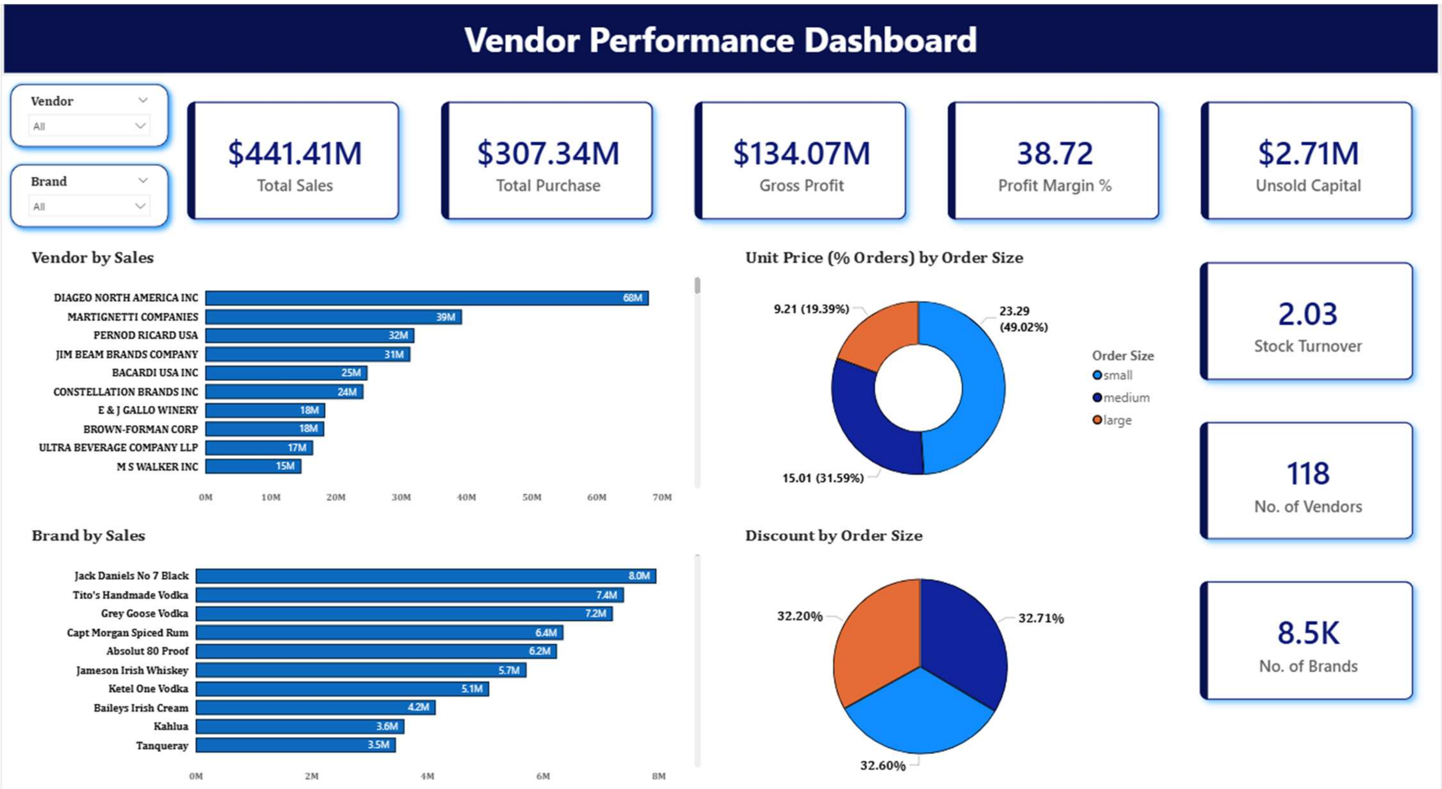
	vendorname	avg_stock_turnover
0	ALISA CARR BEVERAGES	0.62
1	HIGHLAND WINE MERCHANTS LLC	0.71
2	PARK STREET IMPORTS LLC	0.75
3	Circa Wines	0.76
4	KLIN SPIRITS LLC	0.76
5	CENTEUR IMPORTS LLC	0.77
6	Dunn Wine Brokers	0.77
7	SMOKY QUARTZ DISTILLERY LLC	0.78
8	TAMWORTH DISTILLING	0.80
9	THE IMPORTED GRAPE LLC	0.81

6. **Unsold Capital by Vendor**
   Capital locked in unsold inventory per vendor and vendor contribution to total unsold capital.

	vendorname	total_locked_capital	org_locked_capital	contribution_perc
0	DIAGEO NORTH AMERICA INC	722209.05	3953635.53	18.27
1	JIM BEAM BRANDS COMPANY	554665.63	3953635.53	14.03
2	PERNOD RICARD USA	470625.61	3953635.53	11.90
3	WILLIAM GRANT & SONS INC	401960.83	3953635.53	10.17
4	E & J GALLO WINERY	228282.61	3953635.53	5.77
5	SAZERAC CO INC	198436.41	3953635.53	5.02
6	BROWN-FORMAN CORP	177733.74	3953635.53	4.50
7	CONSTELLATION BRANDS INC	133617.62	3953635.53	3.38
8	MOET HENNESSY USA INC	126477.70	3953635.53	3.20
9	REMY COINTREAU USA INC	118598.15	3953635.53	3.00

Power BI Dashboard

Finally, I built an interactive Power BI Dashboard to present insights visually.



Business Impact

1. Helps prioritize high-performing vendors
2. Enables early identification of risky vendors
3. Supports data-driven procurement decisions
4. Improves financial planning and risk mitigation

Recommendations

1. Monitor vendors moving into higher risk buckets
2. Strengthen partnerships with stable ROI vendors
3. Investigate declining segments for inefficiencies
4. Implement automated monitoring dashboards

Tech Stack

1. **SQL** (Advanced queries, aggregations)
2. **Python** (Pandas, EDA)
3. **Data Visualization** tools (Power BI)

This project demonstrates strong analytical capabilities in SQL and Python, along with the ability to translate data into actionable business insights.